

Homework rules

- Group discussion is *highly recommended*. But you have to write down your solution by *yourself*. Aim for clarity and conciseness in your solution, emphasizing the main ideas over low-level details.
- You should *cite* all the references that help you to solve the homework, including the people you have discussed with and the online material you looked at.
- You are required to type your solution using latex and submit a *pdf* to Gradescope and a *tex* file to NTUCool. You may type your solution in English (preferable) or Chinese.
- You are not expected to solve *all* problems. Just try your best! If you don't solve a problem, try your best to write down your thinking process.
- You may use AI however you wish to deepen your understanding of the lecture material. Upload the notes, talk to your AI about them, ask for more explanation or examples; it's all fine. You may not use LLMs in any way to work on your homework. You may not upload assignments, ask for hints, ask how certain concepts from the lectures might be applied to specific homework problems, or upload your assignments to check for correctness or clarity or anything else. You may not include any AI generated content whatsoever in your homework submissions (This policy is taken from MIT 6.5620).
- We will use a *random-checking policy* in which students may be asked to present their homework solutions during TA office hours, with the expectation that everyone will present *at least once*. Evaluation is based on *consistency*, meaning students must be able to reproduce the solutions they originally submitted. If a student fails this consistency check, they will receive a significant *percentage deduction* from their homework score.
- 作業總分為42分，可得最大計分為35分。
- Deadline 4/5(日)23:59，遲交時間最晚4/7(二) 23:59 (遲交成績打9折)，超過則以0分計算。

1. (8 points) **(Uniformity)** Recall that a random variable X over a finite set S is called *uniformly random* if $p_X(x) = \frac{1}{|S|}$ for all $x \in S$. The following problems demonstrate some interplay between independence and uniform randomness. A group $(G, \cdot)$ is a set G endowed with a binary operation $\cdot : G \times G \rightarrow G$, where we will simply write $g_1 \cdot g_2$ for $\cdot(g_1, g_2)$, satisfying the following three properties:

- For any $(g_1, g_2, g_3) \in G^3$, we have that $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$.
- There exists $e \in G$ such that $g = g \cdot e = e \cdot g$ for any $g \in G$.
- For any $g \in G$ there exists $h \in G$ such that $g \cdot h = e = h \cdot g$.

(i) (2 point) Check that $(\{0, 1\}^n, \oplus)$, where $n \in \mathbb{N}$ and $\oplus$ denotes the bitwise exclusive-or operation on $\{0, 1\}^n$, forms a group.

Let U and V be random variables over G . Note that the group operation $\cdot$ induces the random variable $\cdot(U, V)$ over G , which will be denoted by $U \cdot V$. Suppose that U is uniformly random.

- (ii) (2 point) Suppose that U and V are independent. Prove that the random variable $U \cdot V$ is uniformly random.
- (iii) (2 point) Show that U and V may not be independent even if both V and $U \cdot V$ are uniformly random. (Hint: $(G, \cdot) = (\mathbb{Z}_3, +)$ and $V = U$.)
- (iv) (2 point) Show that in the case where $(G, \cdot) = (\mathbb{Z}_2, +)$, the condition that $U \cdot V$ is uniformly random implies that U and V are independent. (Hint: Treat $p_{U,V}(a, b)$, $(a, b) \in \mathbb{Z}_2^2$, as four unknowns, and find sufficiently many equations to conclude that $p_{U,V}(a, b) = \frac{1}{2}p_V(b)$ for all (a, b) .)

2. (2 points) **(Markov)** In the beginning of our proof on Goldreich-Levin Theorem, we quantified a good set G of x 's in which the adversary $\mathbf{Adv}$ has high win rate, and then showed that the size of G is not too small. This is a common technique on dealing with probabilistic arguments. You are now asked to formally show that this technique is valid.

Let $\mathcal{A}$ be a probabilistic algorithm with n bit input such that

$$\Pr[x \xleftarrow{\$} \{0,1\}^n : A(x) = 1] \geq p$$

- . Prove that for any $0 < \epsilon < p$, we have the size of the “good set” $|G_\epsilon| \geq \epsilon 2^n$, where

$$G_\epsilon := \{x \in \{0,1\}^n \mid \Pr[A(x) = 1] \geq p - \epsilon\}$$

You should be able to prove this claim via simple probabilistic operations.

3. (5 points) **(Fun with PRGs)** For each of the following functions G , prove the statement: “if G_1 is a pseudorandom generator, then G is a pseudorandom generator” or provide a counterexample to demonstrate that it is not (assuming that PRGs exist).

Note that $|$ denotes concatenation, e.g., $1011|1011 = 10111011$, $\oplus$ denotes bitwise XOR, and $x_{[a:b]}$ denotes the a -th to the b -th bits of the string x . We also use $l(n) > n$ for all $n \in \mathbb{N}$ to denote the stretch of G_1 .

(i) (2 points) $G(x) = G_1(x) \oplus (x|0^{l(n)-n})$, where $|x| = n$.

(ii) (3 points) $G(x) = G_1(\bar{x})|G_1(x)$

4. (6 points) **(Fun with hardcore bits)**

(i) (3 points) Show that if a one-to-one function f has a hardcore predicate, then f is one-way.

(ii) (3 points) Show that the assumption that f is one-to-one above is necessary. I.e., show that there exists a function f that is not one-to-one, has a hardcore predicate, but is not one-way.

5. (10 points) **(Recap on Hardness Amplification)**

In this problem, you would be asked to review and extend the hardness amplification proof. We first give a brief recap to the outline of proof. In this proof, we aim to show that given a (t, ϵ) -one way function f , we can constructed a (t', ϵ') -one way function $f^{(k)}$, where

$$f^{(k)}(x_1, \dots, x_k) = (f(x_1), \dots, f(x_k))$$

We showed that we can achieve an $\epsilon' = [(1 + \delta)\epsilon]^k + \gamma$, which is almost exponentially smaller to ϵ .

The proof of the one-wayness of $f^{(k)}$, not surprisingly, requires reduction. In other words, we must start from an adversary $\mathbf{Adv}_k$ with at least ϵ' (which is smaller) winning rate on inverting $f^{(k)}$, and construct an reduction $\mathcal{R}$, using $\mathbf{Adv}_k$ as a subroutine, and achieve at least ϵ (which is larger) winning rate on inverting f .

The reduction $\mathcal{R}$ is sketched as follow :

- $\mathcal{R}$ receives an challenge $y = f(x)$ for random $x \xleftarrow{\$} \{0,1\}^n$ from the challenger.
- For each $i \in \{1, 2, \dots, k\}$
 - Retry the following procedure for l times:
 - For each $j \neq i$, sample random $x_j \xleftarrow{\$} \{0,1\}^n$ and compute $y_j = f(x_j)$. Also set $y_i = y$.

- Run $\mathbf{Adv}_k$ with input $\vec{y} = (y_1, \dots, y_k)$, obtain output $\vec{x} = (x_1, \dots, x_k)$
- If $f(x_i) = y$, return x_i to the challenger.

In this problem, we will discuss the details on the proof showing that $\mathcal{R}$ is indeed inverts f with probability at least ϵ . If you are not yet familiar with this part of the proof, you should probably review the course video.

- (3 points) **(Tightness)** Show that given an $\mathbf{Adv}_k$ with win rate ϵ^k , no matter how large l is set, the win rate of the given reduction $\mathcal{R}$ cannot always be amplified to larger than ϵ .
(**Hint.** You probably need to “design” an $\mathbf{Adv}_k$ with specific winning pattern)
 - (3 points) **(Achieving Strong One-Way)** In the lecture, we proved a concrete version of hardness amplification. Now try to apply the concrete result and get the asymptotic version. Prove that, given a $\epsilon(n)$ -weak one-way function f , where $1 - \epsilon(n) \geq \frac{1}{p(n)}$ for some polynomial $p(n)$. You can choose a suitable k (probably as a function of n) such that $f^{(k)}$ is strong one-way.
(**Hint.** As always, you should prove by contradiction. Try starting with an $\mathbf{Adv}_k$ with some non-negligible win rate $q(n)$ and get the parameters tuned. Note that you cannot directly set γ to negligible. (Why?))
 - (4 points) **(Twisting the Reductions)** Let us try to change the way we choose i (the starred line in $\mathcal{R}$) in the reduction $\mathcal{R}$. Consider a reduction $\mathcal{R}_{\text{random}}$ which, instead of looping for each i , only sample one random $i \in \{1, \dots, k\}$ to try for the l iterations. Does $\mathcal{R}_{\text{random}}$ give a similar amplification result?
6. (5 points) **(Expansion of a PRG)** Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ be a PRG. For any polynomial $l(\cdot)$, define $G' : \{0, 1\}^n \rightarrow \{0, 1\}^{l(n)}$ as follows:

$$G'(s) = b_1 \cdots b_{l(n)}, \text{ where} \\ X_0 = s \text{ and} \\ X_{i+1} \parallel b_{i+1} = G(X_i).$$

Then G' is a PRG.

7. (6 points) **(Complete the Goldreich-Levin)**

Recall from the proof of the Goldreich-Levin Theorem that we can only assume that there exists a PPT adversary $\mathbf{Adv}^H$ such that

$$\Pr[x \leftarrow \{0, 1\}^* : \mathbf{Adv}^H(1^n, g(x, r)) = h(x, r)] \geq \frac{1}{2} + \epsilon$$

and we construct the reduction $\mathcal{R}$ as follows: for input $(y = f(x), r)$,

- let $m = \text{poly}(\frac{1}{\epsilon}) = \frac{n}{\epsilon^2}$ and $k = \log m$;
- samples $r_1, \dots, r_k \leftarrow \{0, 1\}^n$ and guess b_j (hope for $b_j = \langle x, r_j \rangle$) for each $j \in [k]$;
- for each $I \subseteq [k]$, set $r_I = \bigoplus_{j \in I} r_j$, $b_I = \bigoplus_{j \in I} b_j$ and
- for each $i \in [n]$, set $r_{iI} = r_I \oplus e_i$, where e_i is the i -th standard unit vector;
- independently send (y, r_{iI}) to $\mathbf{Adv}^H$ and $\mathbf{Adv}^H$ returns b'_{iI} ;
- let $c_{iI} = b_I \oplus b'_{iI}$ and set $x'_i = \text{majority}\{c_{iI}\}_{I \neq \emptyset}$;
- outputs $x' = (x'_1, \dots, x'_n)$ together with the input r .

- (i) (2 points) Show that for each $i \in [n]$, $\{r_{iI}\}_{I \neq \emptyset}$ is pairwise independent.
- (ii) (2 points) Deduce from (i) that, conditioning on $b_j = \langle x, r_j \rangle$ for all $j \in [k]$ (i.e. all guesses are correct), then for each $i \in [n]$, $\{c_{iI}\}_{I \neq \emptyset}$ is pairwise independent.
- (iii) (2 points) Finally prove the following inequality by Chebychev inequality

$$\Pr[x'_i \neq x_i \mid x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle] \leq \frac{1}{\epsilon^2 m},$$

where

$$G_1 := \{x \in \{0, 1\}^n \mid \Pr[r \leftarrow \{0, 1\}^n : \mathbf{Adv}^H(f(x), r) = h(x, r)] > \frac{1}{2} + \frac{\epsilon}{2}\}.$$

Chebyshev's Inequality. Let $X_1, \dots, X_m$ be pairwise independent random variables with the same expectation μ and variance σ^2 . Then for every $\delta > 0$,

$$\Pr \left[\left| \frac{\sum_{i=1}^m X_i}{m} - \mu \right| \geq \delta \right] \leq \frac{\sigma^2}{\delta^2 m}.$$